

* Array:

→ An array is a collection of elements of the same data type.

→ allows to store multiple type of data values.

Syntax:

$\text{int}[] \text{array} = \overset{\text{new int}}{[]};$

→ index starts from "0" and size is fixed.

→ elements must be of same type.

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* Advantages :

- code simplicity
- easy access
- sorting / searching

* Types of array :

- 1-D : a single row of elements
- 2-D : array matrix (i, j) (table like format).

* Jagged array :

- It is a 2-D array where each row can have different number of columns

syntax :-

dataType[][] arrayName = new dataType [number of rows][];

↓
column not specified & can be customized.

* String :

- It is a sequence of characters enclosed within `" "`. It is immutable.

→ Important methods :

- 1) length()
- 2) charAt()
- 3) substring()
- 4) equals() / equalsIgnoreCase()
- 5) toLowerCase() / toUpperCase()
- 6) replace()
- 7) split()
- 8) trim()

* StringBuffer :

→ It is a predefined class used to create mutable strings.

→ Used for multi-threaded environment.

→ It is thread-safe (synchronised)

* StringBuilder :

→ It is also used to create mutable strings but it is not synchronised.

→ Not thread-safe.

→ for Single-threaded environment.

* Thread :

→ Smallest unit of a program which can run independently.

→ It allows a program to do multiple tasks at the same time.

* Multithreading :

→ It is the process of executing two or more threads at the same time within a single program.

* Synchronised :

→ It means only one thread is allowed to use a method at a time.

→ It protects the shared data & prevents the modification of two threads at a same time.

eg:-

```
synchronized void print() {  
    Sys.out.println("Hello");  
}
```